

High-Yield Notes — Every Fact With Its Mechanism

HOW TO USE Every grey **WHY** box is the mechanism that lets you rebuild the fact if you blank. Every pink **TRAP** box is a mistake that costs real marks. | **CONSTANTS**
 $h = 6.626 \times 10^{-34} \text{ J s}$ · $c = 3 \times 10^8 \text{ m/s}$ · $e = 1.602 \times 10^{-19} \text{ C}$ · $m_e = 9.11 \times 10^{-31} \text{ kg}$ · $N_A = 6.022 \times 10^{23}$ · $R_H = 109677 \text{ cm}^{-1}$ · $E_1(\text{H}) = -13.6 \text{ eV}$ · $r_1 = 52.9 \text{ pm}$

CHAPTER 1 · SOME BASIC CONCEPTS OF CHEMISTRY

THE MOLE — A COUNTING UNIT

1 mole = 6.022×10^{23} particles. Defined as the number of atoms in exactly **12 g of carbon-12**. Molar mass (g/mol) is numerically equal to atomic/molecular mass in u.

$$n = \frac{m}{M} = \frac{N}{N_A} = \frac{V}{22.7 \text{ L}} = M \times V(L)$$

Molar volume: 22.71 L at STP (273.15 K, **1 bar**) · **22.4 L** at 273.15 K, **1 atm**.

WHY Avogadro's law: equal volumes of *any* gases at the same T and P hold equal numbers of molecules. So one mole of *every* gas takes the same volume — H_2 and SO_2 alike.

TRAP The rationalised NCERT uses 1 bar → 22.71 L. Only use 22.4 L when the question says 1 atm.

THE FIVE LAWS OF CHEMICAL COMBINATION

Law	Who	Statement / example
Conservation of mass	Lavoisier	Matter is neither created nor destroyed in a reaction
Definite proportions	Proust	A given compound always has the same elements in the same fixed ratio by mass
Multiple proportions	Dalton	Two elements forming more than one compound → masses of one combining with a fixed mass of the other are in a simple whole-number ratio . H_2O vs H_2O_2 : for 2 g H, O is 16 g vs 32 g → 1 : 2
Gaseous volumes	Gay-Lussac	Gases combine in simple whole-number ratios by volume
Avogadro's law	Avogadro	Equal volumes, same T and P → equal number of molecules

CONCENTRATION — AND THE TEMPERATURE QUESTION

Term	Formula	Changes with T?
Mass %	$\frac{w_B}{w_A + w_B} \times 100$	No
Mole fraction	$x_B = \frac{n_B}{n_A + n_B}$	No
Molarity	$M = \frac{n_B}{V_{\text{soln}}(\text{L})}$	YES
Molality	$m = \frac{n_B}{w_A(\text{kg})}$	No
ppm	$\frac{w_{\text{solute}}}{w_{\text{soln}}} \times 10^6$	No

WHY Heat a solution and it **expands**: volume rises, moles do not, so molarity **falls**. Mass never changes with temperature, so molality is fixed. That single sentence is the whole 1-marker.

$$M = \frac{10 \times \% \times d}{M_B} \cdot M_1 V_1 = M_2 V_2$$

FORMULAE, SIGNIFICANT FIGURES, LIMITING REAGENT

$$\text{Molecular} = n \times \text{Empirical}, \quad n = \frac{\text{Molar mass}}{\text{Empirical formula mass}}$$

Benzene: EF = CH (13), M = 78 → n = 6 → MF = **C₆H₆**.

Sig figs: non-zero digits count · zeros *between* digits count · *leading* zeros do not · trailing zeros *after a decimal* do. + and – → fewest **decimal places**; × and ÷ → fewest **significant figures**.

Limiting reagent: divide **moles** ÷ **stoichiometric coefficient** for each reactant — the **smallest** value is limiting.

TRAP Never compare raw masses to find the limiting reagent. Always convert to moles first, then divide by the coefficient.

CHAPTER 2 · STRUCTURE OF ATOM

CATHODE RAYS → THE ELECTRON

Low-pressure gas, high voltage. Rays stream from the **cathode (–)**.

- Travel in **straight lines**
- Deflected **towards the positive plate** → carry **negative** charge
- **Properties are independent of the gas and of the electrode material**

WHY If swapping hydrogen for neon for argon changes nothing, the particle cannot belong to any one gas — it must be a **universal constituent of all matter**. That is exactly how Thomson argued the electron is in every atom.

Thomson measured only the ratio by balancing electric against magnetic deflection:

$$e/m = 1.758820 \times 10^{11} \text{ C kg}^{-1}$$

Deflection depends on (i) magnitude of charge, (ii) mass of the particle, (iii) strength of the field.

TRAP Thomson could **not** find e or m separately — only e/m .

MILLIKAN'S OIL DROP → THE CHARGE

Oil droplets are charged by X-ray-ionised air. The field is tuned until a droplet **hangs motionless** — electric force up exactly balances gravity down — which gives the droplet's charge.

WHY Every droplet's charge came out an **integral multiple of $1.602 \times 10^{-19} \text{ C}$** , never a fraction. That proves charge is **quantised** and this is the fundamental unit.

$$m_e = \frac{e}{(e/m)} = \frac{1.602 \times 10^{-19}}{1.758 \times 10^{11}} = 9.1094 \times 10^{-31} \text{ kg}$$

Thomson gave the ratio → Millikan gave the charge → together they gave the mass. That chain is a 3-marker.

BOHR'S MODEL — THE FOUR POSTULATES

1. The electron revolves in certain **fixed circular orbits** called **stationary states**.
2. The energy of an electron in a stationary state **does not change with time**; energy is emitted or absorbed **only** on jumping between orbits.

3. **Angular momentum is quantised:** $mvr = n \frac{h}{2\pi}$

4. **Bohr's frequency rule:** $\Delta E = E_2 - E_1 = h\nu$

$$r_n = 52.9 \frac{n^2}{Z} \text{ pm} \quad E_n = -13.6 \frac{Z^2}{n^2} \text{ eV} = -2.18 \times 10^{-18} \frac{Z^2}{n^2} \text{ J}$$

WHY Why is the energy negative? We *define* $E = 0$ for an electron infinitely far away and free. A bound electron is more stable than a free one, so it must have **less** energy than zero → negative. More negative = more tightly bound. $n = 1$ is the most negative = ground state.

ANODE RAYS — WHY E/M VARIES

Cathode-ray electrons **knock electrons off the gas atoms**, creating positive ions that stream to the cathode.

	Cathode rays	Anode rays
Identity	Electrons	Positive ions of the gas
Mass	Always $9.11 \times 10^{-31} \text{ kg}$	Depends on the gas
Charge	Always $-1.6 \times 10^{-19} \text{ C}$	+1, +2 ... depends how many e^- stripped
e/m	Constant	Varies with the gas

Lightest positive particle = **H^+ = proton**, $1.676 \times 10^{-27} \text{ kg}$ — and therefore the **largest e/m** of any anode ray.

RUTHERFORD'S α -SCATTERING

Observation	Conclusion
Most α pass straight through	Atom is mostly empty space
A small fraction deflect by small angles	Something positive sits inside
~1 in 20 000 rebound at $>90^\circ$	All the positive charge and nearly all the mass is packed into a tiny nucleus

Nucleus $\sim 10^{-15} \text{ m}$, atom $\sim 10^{-10} \text{ m}$ → nucleus is **10^5 times smaller**.

Two drawbacks: (1) an accelerating electron must continuously radiate energy, spiral in and collapse in $\sim 10^{-8} \text{ s}$ — but atoms are stable; (2) it predicts a **continuous** spectrum, whereas atoms give a **line** spectrum. It also says nothing about electron arrangement.

LIMITATIONS OF BOHR'S MODEL

- Fails for **any atom with more than one electron** — it ignores electron-electron repulsion
- Cannot explain the **fine structure** of spectral lines
- Cannot explain the **Zeeman effect** (magnetic field) or the **Stark effect** (electric field)
- Cannot explain **chemical bonding or molecular shapes**
- **Violates Heisenberg's uncertainty principle** — a fixed orbit means exact position *and* exact momentum at once
- Ignores the **wave nature** of the electron

Works only for one-electron species: **H, He^+ , Li^{2+} , Be^{3+}** .

SPECTRA AND THE RYDBERG EQUATION

Emission: bright lines on a dark background. **Absorption:** dark lines on a bright background, at *the same* positions. A line spectrum is an element's **fingerprint**.

$$\bar{\nu} = 109677 \left(\frac{1}{n_1^2} - \frac{1}{n_2^2} \right) \text{ cm}^{-1}$$

Series	n_1	Region
Lyman	1	Ultraviolet
Balmer	2	Visible
Paschen	3	Infrared
Brackett	4	Infrared
Pfund	5	Infrared

Hook: **Lyman** is **Lowest** $n_1 \rightarrow$ biggest jump $\rightarrow$ highest energy $\rightarrow$ UV. **Balmer** is the one you can **Behold** $\rightarrow$ visible.

Number of spectral lines from level n to ground = $n(n-1)/2$.

DUAL NATURE, UNCERTAINTY, ORBITAL

de Broglie: $\lambda = \frac{h}{mv} = \frac{h}{p}$ · Heisenberg: $\Delta x \cdot \Delta p \geq \frac{h}{4\pi}$

WHY A cricket ball has enormous m , so $\lambda \sim 10^{-34}$ m — unmeasurable. An electron's m is tiny, so λ is atom-sized and its wave nature dominates.

Orbit = a fixed circular path (Bohr — wrong). **Orbital** = a three-dimensional region where the **probability of finding the electron is maximum** (~90%).

ψ is the wave function — **no physical meaning on its own**. ψ^2 is the **probability density** — the chance of finding the electron at that point. That is why orbitals are drawn as clouds.

PHOTOELECTRIC EFFECT — ALL FOUR OBSERVATIONS

Observation	Wave theory says	Reality
Below a threshold frequency ν_0 , no electrons come out — however intense the light	Enough intensity should eventually work	It never works. Frequency is a hard gate
Raising intensity $\rightarrow$ more electrons, same KE each	Brighter = more energy each	Only the number changes
Raising frequency $\rightarrow$ higher KE per electron	Frequency irrelevant	It is the only thing that sets KE
No time lag — emission is instantaneous	Energy must accumulate	Instant

$$h\nu = h\nu_0 + \frac{1}{2}m_e v^2 \implies KE = h(\nu - \nu_0)$$

Work function $W_0 = h\nu_0$ = minimum energy to eject an electron from that metal.

WHY Light arrives as **photons** of energy $E = h\nu$. **One photon hits one electron** and dumps all its energy at once. That single-packet picture explains the threshold, the instant emission, and why intensity only changes the count.

THE FOUR QUANTUM NUMBERS

Symbol	Name	Tells you	Allowed values
n	Principal	Size & energy — the shell	1, 2, 3 ...
l	Azimuthal	Shape — the subshell	0 to (n-1)
m_l	Magnetic	Orientation in space	-l to +l $\rightarrow$ (2l+1) values
m_s	Spin	Direction of spin	$+\frac{1}{2}$ or $-\frac{1}{2}$ only

$l = 0 \rightarrow s, 1 \rightarrow p, 2 \rightarrow d, 3 \rightarrow f$

Subshells in shell $n = n$ · orbitals in a subshell = **2l+1** (s 1, p 3, d 5, f 7) · orbitals in shell $n = n^2$ · electrons in shell $n = 2n^2$

TRAP They are **n, l, m_l , m_s** — not m_1 and m_2 . m_l is orientation; m_s is spin. Check validity with $l \leq n-1$ and $|m_l| \leq l$.

Nodes: radial = **$n - l - 1$** · angular = **l** · total = **$n - 1$** . 3p $\rightarrow$ radial 1, angular 1, total 2.

FILLING RULES — AUFBAU, PAULI, HUND

Aufbau: orbitals fill in order of increasing energy, given by the **(n + l) rule** — lower (n+l) first; if tied, lower n first. 4s (4+0=4) fills before 3d (3+2=5).

Pauli exclusion: no two electrons in an atom have all four quantum numbers the same → an orbital holds **at most 2 electrons, with opposite spins**.

Hund's rule of maximum multiplicity: degenerate orbitals are filled **singly first, all with parallel spin**, before any pairing begins.

WHY Hund's rule exists because electrons repel — they stay in separate orbitals as long as there is room.

	Expected	Actual
Cr (24)	[Ar] 3d ⁴ 4s ²	[Ar] 3d⁵ 4s¹
Cu (29)	[Ar] 3d ⁹ 4s ²	[Ar] 3d¹⁰ 4s¹

Reason: exactly half-filled and completely filled subshells are extra stable, because of (a) **symmetrical distribution** of electrons and (b) greater **exchange energy**.

TRAP 4s fills **before** 3d, but electrons are **removed from 4s first**. Fe = [Ar]3d⁶4s² → Fe²⁺ = [Ar]3d⁶ → Fe³⁺ = [Ar]3d⁵.

TRAP Cr and Cu are **electronic-configuration anomalies**, **not** ionisation-enthalpy anomalies. The IE anomalies are **Be > B** and **N > O**.

THE ENGINE BEHIND EVERY TREND

Every single trend in this chapter is a tug-of-war between three things:

1. **Effective nuclear charge (Z_{eff})** — the net pull the outer electron actually feels. Pulls inward.
2. **Shielding / screening** — inner electrons blocking that pull. Pushes outward.
3. **Number of shells** — more shells, bigger atom.

Learn this one paragraph and you can derive every trend below instead of memorising five of them.

ATOMIC AND IONIC RADIUS

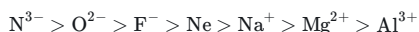
Across a period → DECREASES. Protons are added one at a time but the electrons go into the **same shell**, so shielding barely changes. Z_{eff} rises → the shell is pulled in tighter.

Down a group → INCREASES. A whole **new shell** is added; the extra shielding outweighs the extra nuclear charge.

Cation < parent atom — electrons lost (often a whole shell), the same nuclear charge now holds fewer electrons.

Anion > parent atom — electrons added, more mutual repulsion, the same nuclear charge spread thinner.

Isoelectronic species (same electron count): radius **falls as Z rises**.



All have 10 electrons; Z runs 7→13. More protons, same electrons → smaller.

PERIODIC LAW, PERIODS, BLOCKS

Modern periodic law: the properties of elements are a **periodic function of their atomic numbers**. Mendeleev used atomic *mass*, which gave anomalies (Ar/K, Co/Ni); **Moseley's X-ray work** established atomic *number* as fundamental.

Period	Orbitals filled	Elements
1	1s	2
2	2s 2p	8
3	3s 3p	8
4	4s 3d 4p	18
5	5s 4d 5p	18
6	6s 4f 5d 6p	32
7	7s 5f 6d 7p	32

Block	Groups	Configuration
s	1, 2	ns^{1-2}
p	13-18	$ns^2 np^{1-6}$
d	3-12	$(n-1)d^{1-10} ns^{0-2}$
f	Lanthanoids, actinoids	$(n-2)f^{1-14} (n-1)d^{0-1} ns^2$

TRAP ns^{1-2} means "1 **to** 2 electrons in the ns orbital". It is a **range**, not a subtraction.

IONISATION ENTHALPY — AND BOTH ANOMALIES

Energy **required** to remove the most loosely bound electron from an **isolated gaseous atom in its ground state**. Always **positive** (endothermic).

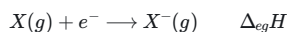
Increases across a period · decreases down a group.

Why $IE_2 > IE_1$ always: after the first electron leaves, the species is a **cation** — the same protons now hold fewer electrons, so every one left is gripped harder.

Anomaly	Reason
Be (899) > B (801)	In Be the electron leaves a 2s orbital. In B it leaves a 2p orbital — higher in energy and shielded by the 2s pair → easier to remove.
N (1402) > O (1314)	N is $2p^3$ — exactly half-filled , extra stable, every electron in its own orbital. O is $2p^4$ — one orbital now holds a pair , and their mutual repulsion makes one easier to pull out.

Same logic repeats: **Mg > Al** and **P > S**.

ELECTRON GAIN ENTHALPY — IN FULL



The atom must be **gaseous** (isolated, no neighbours), **neutral** and in its **ground state**.

Sign convention: energy **released** $\rightarrow \Delta_{eg}H$ is **negative** (the atom wants the electron). Energy must be **supplied** $\rightarrow \Delta_{eg}H$ is **positive** (the atom resists it).

Species	Sign	Why
Halogens	Most negative	One electron short of a full octet
Noble gases	Positive	Octet already complete — the electron must be forced into a new shell
Be, Mg (group 2)	Positive	Stable filled ns^2 subshell
N	~zero	Stable half-filled $2p^3$

Trends: across a period $\rightarrow$ **more negative** (smaller atom, higher Z_{eff} , keener to accept). Down a group $\rightarrow$ **less negative** (bigger atom, the incoming electron sits further from the nucleus).

WHY The anomaly: Cl (–349) is more negative than F (–328) — and S more than O. F and O are **second-period** atoms: very small, with **compact 2p subshells**. Forcing an extra electron into that crowded little region causes **strong inter-electronic repulsion**, which cancels much of the energy released. Cl and S are third-period — roomier, less repulsion, so more energy is actually given out.

Electron affinity vs $\Delta_{eg}H$: electron affinity is the energy *released* (positive when released); $\Delta_{eg}H$ is the enthalpy *change* (negative when released). Opposite signs for the same event. NCERT uses $\Delta_{eg}H$.

ELECTRONEGATIVITY, METALLIC CHARACTER, OXIDES

Electronegativity: the tendency of an atom **in a chemical compound** to attract the **shared pair** of electrons. **Cannot be measured directly** — relative scale, **no units**. Pauling: **F = 4.0 highest**. Increases across a period, decreases down a group.

TRAP Electronegativity applies to an atom **inside a bond**; $\Delta_{eg}H$ applies to an **isolated gaseous atom** and is a measurable energy. Do not swap them.

Metallic character tracks how easily an atom **loses** electrons $\rightarrow$ inversely follows IE. Decreases across a period, increases down a group.

Oxides across period 3: Na_2O , MgO **basic** $\rightarrow Al_2O_3$ **amphoteric** $\rightarrow SiO_2$, P_4O_{10} , SO_3 , Cl_2O_7 **acidic**. Amphoteric = reacts with **both** acids and bases.

Diagonal relationship: Li-Mg, Be-Al, B-Si — the first element of a group resembles the **second element of the next group**, because they end up with similar size and polarising power.

TYPES OF CHEMICAL BOND

Bond	How it forms	Example
Ionic (electrovalent)	Complete transfer of electrons; metal + non-metal; large electronegativity difference	NaCl, MgO
Covalent	Sharing of electron pairs	H ₂ , CH ₄
— non-polar covalent	Shared equally (identical atoms)	H ₂ , Cl ₂ , O ₂
— polar covalent	Shared unequally (different electronegativity)	HCl, H ₂ O
Coordinate (dative)	Both shared electrons come from the same atom	NH ₄ ⁺ , O ₃ , CO
Hydrogen bond	δ+ H attracted to a lone pair on F, O or N	H ₂ O, HF

Favourable conditions for an ionic bond: low ionisation enthalpy of the metal, highly negative electron gain enthalpy of the non-metal, and **high lattice enthalpy**.

Lattice enthalpy = the energy required to completely separate **one mole** of a solid ionic compound into its **gaseous ions**. NaCl = 788 kJ mol⁻¹.

OVERLAPPING OF ATOMIC ORBITALS

A covalent bond forms when two half-filled atomic orbitals **overlap**. **The greater the overlap, the stronger the bond.**

Sign of overlap	Phases	Result
Positive	Same phase (same sign of ψ)	Electron density builds between the nuclei → bond forms
Negative	Opposite phase	Density cancels between the nuclei → antibonding
Zero	Perpendicular arrangement	Constructive and destructive parts cancel exactly → no bond

σ bond — head-on / axial / end-to-end overlap along the internuclear axis:

- **s-s** → H₂
- **s-p** → HF, HCl
- **p-p (axial, e.g. p_z-p_z)** → F₂, Cl₂
- **hybrid-s** or **hybrid-hybrid** → CH₄, C₂H₆

π bond — sideways / lateral overlap of unhybridised p orbitals, perpendicular to the internuclear axis (p_x-p_x, p_y-p_y). The electron cloud sits **above and below** the axis, never on it.

	σ (sigma)	π (pi)
Overlap	Head-on / axial	Sideways / lateral
Extent of overlap	Large	Small
Strength	Stronger	Weaker
Rotation about the bond	Free (cylindrically symmetric)	Restricted — rotating would break the overlap
Formed from	Hybrid orbitals	Unhybridised p orbitals
Exists alone?	Yes	No — only alongside a σ

The rule that unlocks every counting question: single = 1σ · double = 1σ + 1π · triple = 1σ + 2π. The **first** bond of any multiple bond is always σ.

MOLECULAR ORBITAL THEORY — THE BONDING / ANTIBONDING RELATIONSHIP

Two atomic orbitals combine by **LCAO** (linear combination of atomic orbitals). **Two AOs in → always exactly two MOs out**. The total number of MOs always equals the total number of AOs combined.

	Bonding MO (σ, π)	Antibonding MO (σ^*, π^*)
Formed by	Constructive interference — waves in the same phase add	Destructive interference — waves in opposite phase subtract
Wave function	$\psi_b = \psi_A + \psi_B$	$\psi_a = \psi_A - \psi_B$
Electron density between the nuclei	Concentrated there	Zero — there is a NODE between the nuclei
Effect on the nuclei	The shared cloud screens the two nuclei from each other and pulls both inward	The nuclei are exposed to each other and repel
Energy vs the parent AOs	LOWER → stabilising	HIGHER → destabilising
Filled	First (Aufbau still applies)	Only after the bonding MO is full

Five relationships to state in an answer:

1. They are **always produced as a pair** — one bonding and one antibonding from every pair of combining AOs.
2. The antibonding MO is **raised more** than the bonding MO is lowered.

WHY This is exactly why **He₂ does not exist**: 2 electrons in $\sigma 1s$ and 2 in $\sigma^* 1s$ give bond order 0, and the net effect is slightly *destabilising*.

3. Each pair of electrons in an antibonding MO **cancels** one pair in the corresponding bonding MO.
4. Conditions for two AOs to combine: **comparable energy**, **same symmetry** about the molecular axis, and **sufficient overlap**.
5. Bond order = $\frac{N_b - N_a}{2}$. **BO > 0 → the molecule exists. BO = 0 → it does not.**

Energy order — and there are two of them. For **B₂, C₂, N₂** (14 electrons or fewer):

$$\sigma 1s < \sigma^* 1s < \sigma 2s < \sigma^* 2s < \pi 2p_x = \pi 2p_y < \sigma 2p_z < \pi^* 2p_x = \pi^* 2p_y < \sigma^* 2p_z$$

For **O₂, F₂** (more than 14 electrons) the **$\sigma 2p_z$ drops below the π orbitals**:

$$\sigma 1s < \sigma^* 1s < \sigma 2s < \sigma^* 2s < \sigma 2p_z < \pi 2p_x = \pi 2p_y < \pi^* 2p_x = \pi^* 2p_y < \sigma^* 2p_z$$

N₂: $KK \sigma 2s^2 \sigma^* 2s^2 \pi 2p_x^2 \pi 2p_y^2 \sigma 2p_z^2 \rightarrow BO = \frac{10-4}{2} = 3$, all paired → **diamagnetic**

O₂: $KK \sigma 2s^2 \sigma^* 2s^2 \sigma 2p_z^2 \pi 2p_x^2 \pi 2p_y^2 \pi^* 2p_x^1 \pi^* 2p_y^1 \rightarrow BO = \frac{10-6}{2} = 2$, **two unpaired** → **PARAMAGNETIC**

Magnetic behaviour: any **unpaired** electron in any MO → **paramagnetic** (attracted by a magnetic field). All paired → **diamagnetic** (weakly repelled). MOT is the **only** theory that predicts O₂'s paramagnetism — that is its headline success.

Species	Electrons	Bond order	Bond length	Magnetic
O ₂ ⁺	15	2.5	Shortest	Para
O ₂	16	2.0	↓	Para
O ₂ [−]	17	1.5	↓	Para
O ₂ ^{2−}	18	1.0	Longest	Dia

Bond order ↑ ⇒ bond length ↓, bond enthalpy ↑, stability ↑

Isoelectronic species have the same bond order: N₂, CO and NO⁺ all have 14 electrons and bond order 3.

LEWIS STRUCTURES AND FORMAL CHARGE

1. Count **total valence electrons** (add for a negative charge, subtract for a positive one)
2. Put the **least electronegative** atom in the centre — H is **never** central
3. Join each outer atom with a single bond
4. Complete the octets of the **outer** atoms first
5. Leftover electrons → lone pairs on the central atom
6. Still short → convert lone pairs into **double or triple bonds**

Valence electrons: H 1 · Be 2 · B 3 · C 4 · N 5 · O 6 · F/Cl 7 · P 5 · S 6 · Xe 8

$$FC = V - L - \frac{1}{2}B$$

V = valence electrons in the free atom, L = lone-pair electrons, B = bonding electrons. **The best Lewis structure is the one with the smallest formal charges.**

LIMITATIONS OF THE OCTET RULE

Category	Examples
Incomplete octet — central atom has fewer than 8 (elements with fewer than 4 valence electrons)	LiCl, BeH ₂ , BCl₃ / BF₃ (B has 6) , AlCl ₃
Odd-electron molecules — total electrons odd, so the octet cannot be satisfied for every atom	NO, NO₂
Expanded octet — 3rd period onwards have empty d orbitals available	PF₅ (10), SF₆ (12) , H ₂ SO ₄ , PCl ₅

Three further limitations (the forgotten 1-markers):

- It says nothing about the **shape** of a molecule
- It says nothing about the **energy or relative stability** of a molecule
- Some **noble gases form compounds** (XeF₂, XeF₄, KrF₂) despite complete octets

RESONANCE

Definition: when a single Lewis structure cannot describe a molecule accurately, several structures with **similar energy, the same positions of nuclei**, and the same number of bonding and non-bonding electron pairs are taken as the **canonical structures**, and the molecule is described by their **resonance hybrid**.

Molecule	Canonical forms	Evidence
O_3	2	Both O-O bonds equal at 128 pm — between single (148) and double (121)
CO_3^{2-}	3	All three C-O bonds identical
CO_2	3	Both C-O bonds identical

The **hybrid is lower in energy (more stable)** than any canonical form — the difference is the **resonance stabilisation energy**.

TRAP The four things NCERT says resonance is NOT (a favourite question): canonical forms have **no real existence**; the molecule does **not** spend part of its time in each form; there is **no equilibrium** between them; the molecule has **one single structure** which simply cannot be drawn with one Lewis diagram.

VSEPR — THE FULL TABLE

The rule: electron pairs around the central atom get as far apart as possible. Repulsion order:

$lp-lp > lp-bp > bp-bp$

BP	LP	Shape	Example	Angle
2	0	Linear	$BeCl_2$	180°
3	0	Trigonal planar	BF_3	120°
4	0	Tetrahedral	CH_4	109.5°
3	1	Trigonal pyramidal	NH_3	107°
2	2	Bent / V-shaped	H_2O	104.5°
5	0	Trigonal bipyramidal	PCl_5	$120^\circ, 90^\circ$
4	1	See-saw	SF_4	—
3	2	T-shaped	ClF_3	—
6	0	Octahedral	SF_6	90°
5	1	Square pyramidal	BrF_5	—
4	2	Square planar	XeF_4	90°

WHY The bond angle answer: CH_4 $109.5^\circ \rightarrow NH_3$ $107^\circ \rightarrow H_2O$ 104.5° . All three have **four** electron pairs, so all start tetrahedral. NH_3 has **one lone pair**, H_2O has **two**. A lone pair is held by only one nucleus, so it spreads out wider and **squeezes the bonding pairs closer**. More lone pairs $\rightarrow$ smaller angle.

TRAP SF_4 is see-saw. XeF_4 is SQUARE PLANAR. Do not swap them.

HYBRIDISATION — THE THREE STEPS AND THE SHORTCUT

1 Promotion — an electron is promoted to an empty orbital of the same shell to create enough half-filled orbitals. **2 Mixing** — orbitals of *nearly equal energy* mix to give an equal number of **identical hybrid orbitals**. **3 Overlap** — each hybrid overlaps head-on with another orbital → **σ bond**.

Steric number = (atoms bonded to the central atom) + (lone pairs on it)

SN	Hybridisation	Geometry
2	sp	Linear, 180°
3	sp ²	Trigonal planar, 120°
4	sp ³	Tetrahedral, 109.5°
5	sp ³ d	Trigonal bipyramidal
6	sp ³ d ²	Octahedral, 90°

$$\text{Lone pairs} = \frac{(\text{valence } e^- \text{ of central atom}) - (\text{number of bonds formed})}{2}$$

H₂O: O has 6, forms 2 bonds → LP = 2 → SN = 4 → **sp³**, **bent**.

NH₃: N has 5, forms 3 bonds → LP = 1 → SN = 4 → **sp³**, **trigonal pyramidal**.

CO₂: C has 4, forms 4 bonds → LP = 0 → SN = 2 (only two atoms attached) → **sp**, **LINEAR**.

WHY "sp³" simply names which orbitals were mixed — one s and three p → four identical sp³ hybrids. The superscript is a **count of p orbitals**, not a power.

TRAP Hybrid orbitals only ever make σ bonds. Every π bond comes from a leftover **unhybridised p** orbital.

C₂H₄: sp² C, one p left over → 1σ + 1π. C₂H₂: sp C, two p left over → 1σ + 2π.

DIPOLE MOMENT — POLAR VS NON-POLAR

$$\mu = Q \times r \cdot \text{unit } \mathbf{\text{debye (D)}}, 1 \text{ D} = 3.33564 \times 10^{-30} \text{ C m}$$

μ is a VECTOR — drawn as a crossed arrow pointing **from positive to negative**. For a molecule, μ is the **vector sum of all bond moments**.

$$\mu = 0 \Rightarrow \text{non-polar molecule} \quad \mu \neq 0 \Rightarrow \text{polar molecule}$$

WHY A molecule can have polar bonds and still be non-polar. If the geometry is symmetric, the bond moments cancel as vectors. Rule of thumb: **no lone pairs on the central atom AND all surrounding atoms identical → non-polar**. Lone pairs or mixed atoms break the symmetry and make it polar.

Molecule	Shape	μ / D	Verdict
BeF ₂ , CO ₂	Linear	0	Non-polar
BF ₃	Trigonal planar	0	Non-polar
CCl ₄ , CH ₄	Tetrahedral	0	Non-polar
H ₂ O	Bent	1.85	Polar
NH ₃	Pyramidal	1.47	Polar
NF ₃	Pyramidal	0.23	Weakly polar
CHCl ₃	Tetrahedral	1.04	Polar

HF 1.78 > HCl 1.07 > HBr 0.79 > HI 0.38 — falls as the electronegativity difference falls.

WHY NH₃ vs NF₃. Both pyramidal, both one lone pair on N. In **NH₃**, N is more electronegative than H, so the three N-H bond moments point **towards N — the same side as the lone-pair dipole**. They **add** → large μ. In **NF₃**, F is more electronegative than N, so the N-F moments point **away from N — opposite to the lone-pair dipole**. They **partly cancel** → tiny μ.

BOND PARAMETERS & HYDROGEN BONDING

Bond length — equilibrium distance between two bonded nuclei. **Bond angle** — angle between two bonds at a central atom. **Bond enthalpy** — energy to break one mole of bonds in the gaseous state: H₂ 435.8, O₂ 498, **N₂ 946 kJ mol⁻¹** (why N₂ is so unreactive). **Bond order** — number of bonds between two atoms.

Hydrogen bond: the attractive force binding the hydrogen of one molecule to a **highly electronegative atom — F, O or N**. Strength **10-100 kJ mol⁻¹**.

WHY The large electronegativity difference makes the H-F/O/N bond strongly polar, so H carries **δ+** and is drawn to the lone pair on a **δ-** atom nearby.

- Intermolecular** — between *different* molecules: HF, H₂O, NH₃, alcohols. **Raises** boiling point.
- Intramolecular** — *within* one molecule: **o-nitrophenol**. **Lowers** boiling point, because no chain of molecules can form.

Consequences to quote: abnormally high boiling point of water vs H₂S · **ice is less dense than water** (H-bonds hold an open cage structure) · high surface tension and viscosity · the structure of proteins and DNA.

WHAT AN OXIDATION NUMBER ACTUALLY IS

Oxidation number is a bookkeeping device. It is the charge an atom **would** carry if every bond in the compound were **100% ionic** — that is, if in every bond you handed **both** shared electrons entirely to the **more electronegative** atom.

WHY It is not the real charge. In H_2O oxygen is assigned -2 , but its actual partial charge is only about -0.4 — the bond is polar covalent, not ionic. We *pretend* it is fully ionic so that electron movement can be tracked with whole numbers.

Why it was invented: in a covalent reaction nobody transfers a whole electron, so "loss of electrons" has no literal meaning. The oxidation number gives an unambiguous way to say whether an atom **gained or lost control** of electron density.

Where the rules come from: in H-Cl , Cl is more electronegative, so it takes both electrons $\rightarrow \text{Cl} = -1$, $\text{H} = +1$. In Cl-Cl the two atoms are identical, so the pair is split evenly $\rightarrow$ each = 0. **That is exactly why every free element is 0.** Every rule below is just a consequence of relative electronegativities.

THE SEVEN RULES, IN PRIORITY ORDER

- Free element = 0** — Na , O_2 , P_4 , S_8 , Cl_2
- Monatomic ion** = its charge — $\text{Na}^+ = +1$, $\text{Cl}^- = -1$
- F is always -1** (most electronegative element — it can never be positive)
- O is -2** , except: **-1 in peroxides** (H_2O_2 , Na_2O_2), **$-\frac{1}{2}$ in superoxides** (KO_2), **$+2$ in OF_2**
- H is $+1$** , except **-1 in metal hydrides** (NaH , CaH_2)
- Group 1 = **$+1$** , Group 2 = **$+2$**
- Sum = 0** for a neutral molecule; **sum = the charge** for an ion

TRAP Oxygen is -2 , not $+2$. The only place oxygen is positive is **OF_2 ($+2$)** and O_2F_2 ($+1$) — because fluorine is the one element more electronegative than oxygen.

Method: let the unknown be x , apply rule 7, solve.

KMnO_4 : $(+1) + x + 4(-2) = 0 \rightarrow \text{Mn} = +7$

$\text{Cr}_2\text{O}_7^{2-}$: $2x + 7(-2) = -2 \rightarrow \text{Cr} = +6$

H_2SO_4 : $2(+1) + x + 4(-2) = 0 \rightarrow \text{S} = +6$

OXIDATION, REDUCTION, AND THE AGENTS

	Oxidation	Reduction
Electrons	Loss	Gain
Oxidation number	Increases	Decreases
Oxygen / hydrogen	Gains O, loses H	Loses O, gains H

OIL RIG — Oxidation Is Loss, Reduction Is Gain.

Oxidising agent = **accepts** electrons = is itself **reduced**.

Reducing agent = **donates** electrons = is itself **oxidised**.

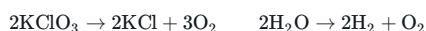
TRAP The agent does the **opposite** of what happens to it. An oxidising agent oxidises *the other substance*, so it itself gets reduced. This single confusion costs more marks than anything else in the chapter.

THE FOUR TYPES OF REDOX REACTION

1 Combination — two species join, at least one an element



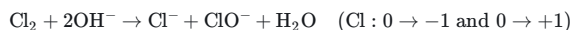
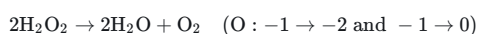
2 Decomposition — one compound splits



3 Displacement — one ion or atom is replaced by another



4 Disproportionation — the same element is **simultaneously oxidised AND reduced**



WHY Requirement for disproportionation: the element must have **at least three oxidation states** and must currently sit in an **intermediate** one — so it has room to move both up and down.

ION-ELECTRON (HALF-REACTION) METHOD — NINE STEPS

- Write the **skeletal ionic equation**
- Assign oxidation numbers; identify what is **oxidised** and what is **reduced**
- Split into **two half-reactions**
- Balance all atoms **except O and H**
- Balance **O** by adding **H_2O** ; balance **H** by adding **H^+**
- Balance **charge** by adding electrons to the more positive side
- Multiply** the half-reactions so the electrons are equal
- Add** them; cancel the electrons
- Basic medium only:** add as many **OH^-** to **BOTH sides** as there are H^+ ; combine $\text{H}^+ + \text{OH}^- \rightarrow \text{H}_2\text{O}$; cancel

TRAP Step 9 is the one people forget. If the question says **alkaline / basic medium** and your final answer still contains H^+ , you have not finished.

PAIRED TERMS — KNOW BOTH HALVES OF EACH

Term	Means	Its partner	Means
Paramagnetic	Has unpaired electrons → attracted by a magnetic field (O_2 , NO)	Diamagnetic	All electrons paired → weakly repelled (N_2 , F_2)
Polar	$\mu \neq 0$ — bond moments do not cancel (H_2O , NH_3 , HCl)	Non-polar	$\mu = 0$ — symmetric, moments cancel (CO_2 , BF_3 , CCl_4)
Bonding MO	Constructive overlap, density between nuclei, lower energy	Antibonding MO	Destructive overlap, node between nuclei, higher energy
σ bond	Head-on overlap, stronger, free rotation	π bond	Sideways overlap, weaker, no rotation
Oxidation	Loss of e^- , oxidation number increases	Reduction	Gain of e^- , oxidation number decreases
Oxidising agent	Accepts e^- , is itself reduced	Reducing agent	Donates e^- , is itself oxidised
Ionisation enthalpy	Energy required to remove an e^- — always positive	Electron gain enthalpy	Enthalpy change on adding an e^- — usually negative
Electronegativity	Attraction for a shared pair, atom in a bond , no units	Electron affinity	Energy released by an isolated gaseous atom, measurable
Intermolecular H-bond	Between molecules → raises b.p. (H_2O , HF)	Intramolecular H-bond	Within one molecule → lowers b.p. (o-nitrophenol)
Empirical formula	Simplest whole-number ratio (CH)	Molecular formula	The actual number of atoms (C_6H_6)
Acidic oxide	Non-metal oxide, reacts with base (SO_3 , Cl_2O_7)	Basic / amphoteric	Metal oxide (Na_2O) / reacts with both (Al_2O_3)
Orbit	Fixed circular path — Bohr's idea, wrong	Orbital	3-D region of maximum probability (~90%)
Shielding	Inner electrons blocking the nuclear pull	Effective nuclear charge	The net pull the outer electron actually feels
Emission spectrum	Bright lines on a dark background	Absorption spectrum	Dark lines on a bright background, same positions
Degenerate orbitals	Orbitals of equal energy (the three 2p)	Non-degenerate	Different energies (2s vs 2p in a multi-electron atom)

THE TWELVE MISTAKES THAT COST THE MOST MARKS

Wrong	Right
Oxygen's oxidation number is +2	-2 ; -1 in peroxides, $-\frac{1}{2}$ in superoxides, +2 only in OF_2
The IE anomalies are Cr and Cu	Cr and Cu are configuration anomalies. The IE anomalies are Be > B and N > O
CO_2 is trigonal planar	Linear, sp — only two atoms attached, SN = 2
XeF_4 is see-saw	XeF_4 is square planar. SF_4 is see-saw
Quantum numbers are n, l, m_l , m_s	n, l, m_l, m_s — m_l is orientation, m_s is spin
ns^{1-2} means "1 minus 2"	It is a range : 1 to 2 electrons
Intensity of light raises the KE of photoelectrons	Intensity raises the number ; only frequency raises KE
Molality changes with temperature	Molarity does (volume expands). Molality never does
Fill 4s first, so empty 4s last	Fill 4s first, but remove from 4s FIRST too
Resonance forms flip back and forth	They have no real existence ; there is one hybrid structure, no equilibrium
An oxidising agent is oxidised	An oxidising agent is reduced
Leaving H^+ in a basic-medium redox answer	Add OH^- to both sides , combine to H_2O , cancel